SVKM's NMIMS (Deemed-to-be University)

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A Machine Learning Group Activity

Literature Review: Research Papers

DemandIQ

Machine Learning for Product Demand and Stockout Risk Prediction

Course: Machine Learning

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Submitted by

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1. Introduction

DemandIQ forecasts store demand, predicts the risk of a stockout, recommends how much to order and explains every number. Before building it, we studied the problem from two kinds of sources: 20 reference books for the foundations (forecasting practice, inventory theory, machine learning, explainability and causal reasoning) and 15 peer-reviewed research papers, most from 2019-2026, to see what current systems achieve and where they stop.

This document lists the **15 research papers**. Each paper is compared in Table 1, its role in DemandIQ is summarised in Table 2, and its full IEEE reference with a **clickable DOI link** is given in Section 4. The 20 reference books are listed in Section 5. Reference numbers are the same as in the project report, so they can be matched directly.

Blue underlined text is a link: click it to open the paper on the publisher's website.

2. Comparison of Reviewed Research Papers

Table 1. Comparison of reviewed research papers

Ref.	Author / Year	Technique	Dataset	Metrics / Results	Limitations
[25]	Seyedan & Mafakheri, 2020	Survey: time series, clustering, KNN, NN, SVR	Review of published studies	Taxonomy of big-data forecasting methods for supply chains	No empirical comparison; no stock decision
[3]	Makridakis et al., 2022	M5 competition; mostly LightGBM ensembles won	42,840 Walmart series with prices and calendar	ML methods beat statistical benchmarks	Accuracy only; large ensembles hard to explain
[26]	Ke et al., 2017	Gradient boosting with GOSS and EFB	Public benchmark datasets	Up to 20x faster training at similar accuracy	Algorithm paper; no retail evaluation
[27]	Gür Ali et al., 2009	Regression trees with promotion features	SKU-store sales, European grocer	Big gains during promotions; pooling SKUs helps	Older data; no inventory link
[28]	Huber & Stuckenschmidt, 2020	ML vs established methods; special-day features	Daily sales of a bakery chain	ML wins; calendar special days matter	One retailer; forecast only
[29]	Spiliotis et al., 2022	Croston-type vs ML with cross-learning	Daily intermittent SKU demand	Some ML beat Croston in accuracy and bias	No explanation; accuracy only
[30]	Lim et al., 2021	Temporal Fusion Transformer (deep learning)	Retail, electricity, traffic sets	About 7% lower P50 and 9% lower P90 loss	Heavy model, GPU; hard for managers
[31]	Liu et al., 2025	Classical ML classifiers, class imbalance	1.6 M+ retailer SKUs	Stock level, 3-month forecast, recent sales are top predictors	Global importance only; no order size
[32]	Theodorou et al., 2025	Forecasts + inventory simulation	M5 Walmart data	Accuracy-cost link can be weak	Simulated stock; no explanation
[33]	Bertsimas & Kallus, 2020	Prescriptive analytics (kNN, trees, kernels)	Media distributor, ~1 bn units/yr	Data-driven decisions beat predict-then-optimise	Mathematically heavy to audit
[34]	Ban & Rudin, 2019	Feature-based (big data) newsvendor	Hospital nurse staffing	23-24% lower cost than benchmark	Single period; not retail stores
[35]	Lundberg & Lee, 2017	SHAP additive feature attribution	Benchmark models	Unique, consistent per-prediction attributions	Describes the model, not causality
[36]	Mishra et al., 2026	LightGBM + PFI, PDP, SHAP, ICE	UCI Online Retail II, 41 features	RMSE 33.71, R^2 0.243 (SKU-day)	Explanations stop at forecast

Ref.	Author / Year	Technique	Dataset	Metrics / Results	Limitations
[37]	Rožanec et al., 2022	Knowledge-graph enriched XAI	Real demand-forecasting case	Links drivers to news and external data	Complex set-up; no stock decision
[38]	Chernozhukov et al., 2018	Double/debiased ML, cross-fitting	Econometric examples	Valid effect estimates where naive ML is biased	Needs causal design; not a forecaster

3. How Each Paper Shaped DemandIQ

Table 2. Role of each paper in the project, with a link to open it

Ref.	Author / Year	How it shaped DemandIQ	Open
[25]	Seyedan & Mafakheri, 2020	Showed the move from classical time-series models to data-driven methods for supply-chain demand.	Open paper
[3]	Makridakis et al., 2022	ML methods, mostly LightGBM, beat statistical benchmarks on Walmart data, so we chose tree-based models.	Open paper
[26]	Ke et al., 2017	LightGBM trains much faster than older boosting at similar accuracy: gradient-boosted trees as the workhorse.	Open paper
[27]	Gür Ali et al., 2009	Promotions need explicit features, so the model has promotion inputs.	Open paper
[28]	Huber & Stuckenschmidt, 2020	Calendar special days behave differently from normal days, so we added holiday and closure features.	Open paper
[29]	Spiliotis et al., 2022	Training one model across many series (cross-learning) helps, so one model forecasts all 935 stores.	Open paper
[30]	Lim et al., 2021	Accurate but heavy deep model; the course brief was classical ML, so it was not used.	Open paper
[31]	Liu et al., 2025	Stock level and near-term forecast are the strongest predictors, so the risk model sees both.	Open paper
[32]	Theodorou et al., 2025	The most accurate forecast is not always the cheapest to stock, so we judge policies on stockouts, fill rate and cost.	Open paper
[33]	Bertsimas & Kallus, 2020	Prescriptive methods order directly but are hard to audit, so we keep a readable order-up-to rule.	Open paper
[34]	Ban & Rudin, 2019	Feature-based newsvendor: large savings, but single period; supports linking ML to the order decision.	Open paper
[35]	Lundberg & Lee, 2017	SHAP gives each input a fair share of one prediction; used to explain both models.	Open paper
[36]	Mishra et al., 2026	Closest work to ours (LightGBM + SHAP), but explanations stop at the forecast.	Open paper
[37]	Rožanec et al., 2022	Enriches explanations with external knowledge; again attached to the forecast only.	Open paper
[38]	Chernozhukov et al., 2018	Naive ML effect estimates can be biased, so every what-if is labelled as model behaviour.	Open paper

4. References: Research Papers (IEEE format)

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